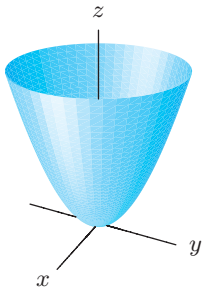

Figure 12.72: Elliptical paraboloid $z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

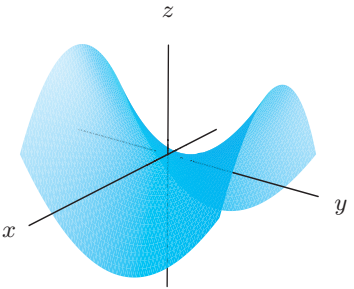

Figure 12.73: Hyperbolic paraboloid $z = -\frac{x^2}{a^2} + \frac{y^2}{b^2}$

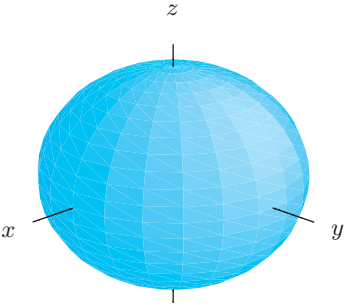

Figure 12.74: Ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

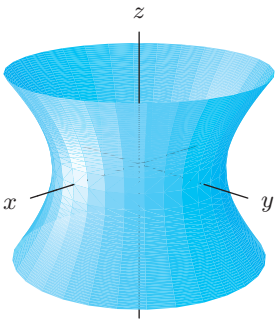

Figure 12.75: Hyperboloid of one sheet $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

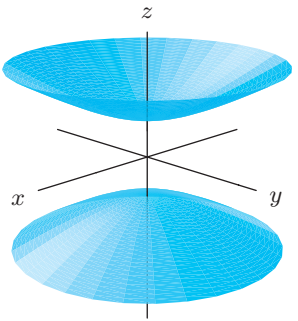

Figure 12.76: Hyperboloid of two sheets $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$

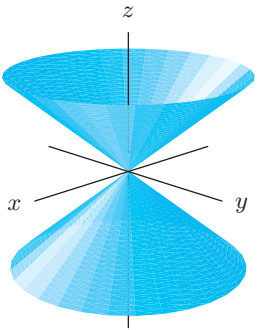

Figure 12.77: Cone $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$

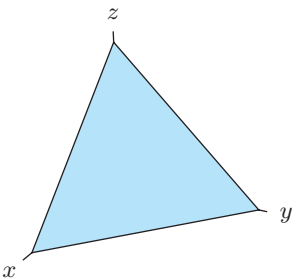

Figure 12.78: Plane $ax + by + cz = d$

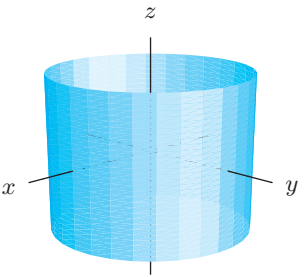

Figure 12.79: Cylindrical surface $x^2 + y^2 = a^2$

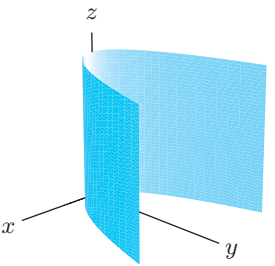

Figure 12.80: Parabolic cylinder $y = ax^2$